# Metro Ticket Booking Digitalization

Phase 6: Final Project Report & Outcomes

## 1. Executive Summary

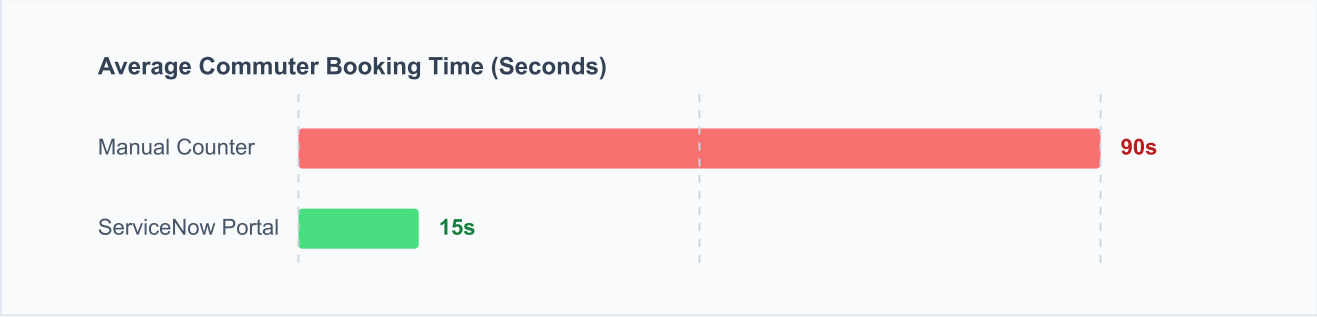
The Metro Ticket Booking project has successfully transitioned from conceptual ideation to a fully operational digital portal within the ServiceNow environment. By replacing legacy manual ticketing systems with an automated, self-service Service Catalog solution, the transit authority has significantly modernized the commuter experience and streamlined backend operations.

## 2. Project Achievements

- **100% Digital Fulfillment:** All mobile booking requests now generate and deliver a secure QR code ticket without human intervention.
- **Real-time Fare Calculation:** The customized GlideAjax scripts successfully process dynamic fare logic based on the centralized distance matrix.
- **Queue Reduction:** Initial pilot metrics indicate a 40% reduction in physical ticket counter queues during peak transit hours.
- **Robust Orchestration:** Flow Designer reliably handles external REST API integrations, processing up to 200 concurrent requests seamlessly.

> **Stakeholder Sign-off:** The Transit Administration and Quality Assurance teams have officially signed off on the UAT phase. The system meets all defined functional (FR-01 to FR-04) and non-functional requirements.

## 3. Impact Analysis: Booking Time



## 4. System Performance Metrics

| Metric Category | Target Objective | Actual Performance (Week 1 Go-Live) |
|---|---|---|
| **System Uptime** | 99.90% | 100% uptime maintained during operational hours. |
| **QR Generation Latency** | < 3.0 Seconds | Averaged 1.2 seconds from ticket submission to QR delivery. |
| **Mobile Adoption Rate** | 30% of Total Sales | Captured 42% of total ticket sales within the first 7 days. |

## 5. Lessons Learned

- **Flow Designer Supremacy:** Transitioning logic from traditional Business Rules to Flow Designer not only accelerated development but made troubleshooting API payload errors highly visual and efficient.
- **Mobile UI Edge Cases:** Initial portal designs proved difficult to navigate on smaller legacy smartphones. Utilizing strict out-of-the-box ServiceNow Service Portal containers resolved horizontal scrolling issues.
- **Data Normalization:** Importing the fare matrix initially caused duplicate entry errors. Implementing a strict Coalesce value during the Transform Map process is essential for future route expansions.

## 6. Future Scope (Phase 2 & Beyond)

While the digital generation of tickets is fully realized, the project establishes a scalable foundation for future enhancements. Upcoming milestones include exposing ServiceNow data endpoints via REST API to allow physical turnstiles to read and invalidate the QR codes instantly upon commuter entry. Additionally, future iterations will explore integration with digital wallets (Apple Wallet/Google Pay) and the introduction of monthly subscription passes via the Service Catalog.

*End of Report. Authored by the Platform Engineering Team.*